WORK, ENERGY AND POWER

Multiple Choice Questions — Practice Problem Set

SECTION 1: MULTIPLE CHOICE QUESTIONS

Q1. Find the angle between the two vectors **P** = 5i^ + 2j^ + 4k^ and **Q** = 2i^ - 5j^ + 0k^.

(A) 30° (B) 45°
(C) 60° (D) 90°

Q2. A vector perpendicular to (i^ + 2j^) and having a magnitude of 3√5 units is:

(A) 3i^ + 6j^ (B) 6i^ - 3j^
(C) 4i^ - 2j^ (D) i^ - 2j^

Q3. The component of vector **A** along **B** is √3 times that of the component of **B** along **A**. The ratio A : B is:

(A) 1 : √3 (B) √3 : 1
(C) 2 : √3 (D) √3 : 2

Q4. If θ is the angle between unit vectors **A** and **B**, then (1 - **A**·**B**) / (1 + **A**·**B**) is equal to:

(A) $\tan^2(\theta/2)$ (B) $\sin^2(\theta/2)$
(C) $\cot^2(\theta/2)$ (D) $\cos^2(\theta/2)$

Q5. A body of mass 10 kg is raised from a certain depth. By the time it is raised by 10 m, its velocity is 2 m/s. Work done during this time is (g = 9.8 m/s²):

(A) 980 J (B) 20 J
(C) 960 J (D) 1000 J

Q6. A bucket filled with water weighing 20 kg is raised from a well of depth 20 m. If the linear density of the rope is 0.2 kg/m, the total work done is (g = 10 m/s²):

(A) 4000 J (B) 4040 J
(C) 4400 J (D) 4200 J

Q7. A uniform chain of length 2 m is kept on a table such that 60 cm hangs freely over the edge. Total mass is 4 kg. Work done in pulling the entire chain onto the table is (g = 10 m/s²):

(A) 7.2 J
(B) 3.6 J
(C) 120 J
(D) 1200 J

Q8. A rectangular block of dimensions 6m × 4m × 2m and density 1.5 g/cm³ lies on horizontal ground with its largest face in contact. Work done in arranging it with its smallest face in contact with ground is (g = 10 m/s²):

(A) 2880 kJ
(B) 1440 kJ
(C) 3800 kJ
(D) 720 kJ

Q9. A body of mass 5 kg moving along a straight line accelerates from 4 m/s to 8 m/s upon application of a 10 N force. Which statement is correct?

(A) Work done = 120 J
(B) Displacement = 12 m
(C) Time for change = 2 s
(D) All of the above

Q10. A uniform rod of length 2 m and mass 5 kg lies on a horizontal surface. Work done in raising one end until the rod makes an angle of 30° with the horizontal is (g = 10 m/s²):

(A) 25 J
(B) 50 J
(C) 25√3 J
(D) 50√3 J

Q11. A force F(x) = (4x² + 3x) N acts on a particle displacing it from x = 2 m to x = 3 m. The work done by the force is:

(A) 32.8 J
(B) 3.28 J
(C) 0.328 J
(D) Zero

Q12. A force of 5 N is applied to a 20 kg mass initially at rest. The work done during the 3rd second of its motion is:

(A) 25/8 J
(B) 25/4 J
(C) 12 J
(D) 25 J

Q13. A spring requires 10 N force to stretch by 1 mm. Work done in stretching the spring through 40 mm is:

(A) 84 J
(B) 68 J
(C) 23 J
(D) 8 J

Q14. A force $F(x) = 7 - 2x + 3x^2$ N acts on a body of mass 2 kg, displacing it from x = 0 to x = 5 m. Work done in joules is:

(A) 70 J
(B) 270 J
(C) 35 J
(D) 135 J

Q15. The work done in lifting a body of mass 20 kg and relative density 3.2 to a height of 8 m in water is ($g = 10\ m/s^2$):

(A) 110 J
(B) 1600 J
(C) 500 J
(D) 1100 J

Q16. A body of mass 3 kg experiences displacement $S = t^3/3$ (in m). Work done by the force in first 2 seconds is:

(A) 2 J
(B) 3.8 J
(C) 5.2 J
(D) 24 J

Q17. Two spheres with radii ratio 1:2 move with velocities ratio 3:4. If their densities ratio is 3:2, the ratio of their kinetic energies is:

(A) 27 : 64
(B) 1 : 1
(C) 27 : 256
(D) 9 : 64

Q18. When a spring is compressed by 3 cm, stored potential energy is U. When compressed by an additional 3 cm, the increase in potential energy is:

(A) 4 U
(B) U
(C) 2 U
(D) 3 U

Q19. If the kinetic energy of a body increases by 125%, the percentage increase in its momentum is:

(A) 50%
(B) 62.5%
(C) 250%
(D) 200%

Q20. The kinetic energy of a body is K. If one-fourth of its mass is removed and its velocity is doubled, its new kinetic energy is:

(A) K
(B) 3 K
(C) 4 K
(D) 9/4 K

SECTION 2: ANSWER KEY & DETAILED SOLUTIONS

ANSWER KEY SUMMARY

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
(D)	(B)	(B)	(A)	(D)	(C)	(B)	(B)	(D)	(A)
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
(A)	(A)	(D)	(D)	(D)	(D)	(C)	(D)	(A)	(B)

STEP-BY-STEP SOLUTIONS

Question 1

Correct Answer: (D) 90°

Explanation: Compute the dot product: **P** · **Q** = (5)(2) + (2)(-5) + (4)(0) = 10 - 10 + 0 = 0. Since the dot product is zero, cos θ = 0, giving θ = 90°.

Question 2

Correct Answer: (B) 6i^ - 3j^

Explanation: Checking option (B): Dot product (i^ + 2j^) · (6i^ - 3j^) = (1)(6) + (2)(-3) = 0. Magnitude = $\sqrt{(6^2 + (-3)^2)} = \sqrt{45} = 3\sqrt{5}$ units.

Question 3

Correct Answer: (B) √3 : 1

Explanation: Component of A along B is A cos θ; component of B along A is B cos θ. Given A cos θ = √3 (B cos θ), cancelling cos θ gives A/B = √3/1.

Question 4

Correct Answer: (A) $\tan^2(\theta/2)$

Explanation: Since **A** and **B** are unit vectors, **A**·**B** = cos θ. Thus $(1 - \cos\theta)/(1 + \cos\theta) = [2\sin^2(\theta/2)] / [2\cos^2(\theta/2)] = \tan^2(\theta/2)$.

Question 5

Correct Answer: (D) 1000 J

Explanation: Total work done = Potential Energy gain + Kinetic Energy gain = $mgh + \frac{1}{2}mv^2 = 10(9.8)(10) + \frac{1}{2}(10)(2^2) = 980 + 20 = 1000$ J.

Question 6

Correct Answer: (C) 4400 J

Explanation: Rope mass = 0.2 × 20 = 4 kg. Total work = Work on bucket + Work on rope center of gravity = $m_b g h + m_r g (h/2) = 20(10)(20) + 4(10)(10) = 4000 + 400 = 4400$ J.

Question 7

Correct Answer: (B) 3.6 J

Explanation: Hanging fraction 1/n = 0.6/2 = 3/10 ■ n = 10/3. Work done $W = mgL / (2n^2) = 4(10)(2) / [2(10/3)^2] = 80 / (200/9) = 3.6$ J.

Question 8

Correct Answer: (B) 1440 kJ

Explanation: Volume V = 48 m³, mass m = 48 × 1500 = 72,000 kg. Initial CG height h■ = 2/2 = 1m; final CG height h■ = 6/2 = 3m. W = mg(h■ - h■) = 72000(10)(3 - 1) = 1,440,000 J = 1440 kJ.

Question 9

Correct Answer: (D) All of the above

Explanation: Acceleration a = F/m = 10/5 = 2 m/s². Time t = (v - u)/a = (8 - 4)/2 = 2 s. Displacement s = (v² - u²)/(2a) = (64 - 16)/4 = 12 m. Work W = F × s = 10 × 12 = 120 J. All statements are true.

Question 10

Correct Answer: (A) 25 J

Explanation: Height gained by center of gravity h = (L/2) sin 30° = 1 × 0.5 = 0.5 m. Work done W = mgh = 5(10)(0.5) = 25 J.

Question 11

Correct Answer: (A) 32.8 J

Explanation: W = ∫■³ (4x² + 3x) dx = [(4/3)x³ + (3/2)x²]■³ = (4/3)(27 - 8) + (3/2)(9 - 4) = 76/3 + 15/2 = 25.33 + 7.5 = 32.83 J.

Question 12

Correct Answer: (A) 25/8 J

Explanation: Acceleration a = F/m = 5/20 = 0.25 m/s². Distance in 3rd second S■ = ½a(2n - 1) = ½(0.25)(5) = 5/8 m. Work W = F × S■ = 5 × (5/8) = 25/8 J.

Question 13

Correct Answer: (D) 8 J

Explanation: Spring constant k = F/x = 10 / (10■³ m) = 10■ N/m. Work W = ½kx² = ½(10■)(40 × 10■³)² = ½(10■)(1.6 × 10■³) = 8 J.

Question 14

Correct Answer: (D) 135 J

Explanation: W = ∫■■ (7 - 2x + 3x²) dx = [7x - x² + x³]■■ = 35 - 25 + 125 = 135 J.

Question 15

Correct Answer: (D) 1100 J

Explanation: Apparent force F = mg(1 - d_water / d_body) = 20(10)(1 - 1/3.2) = 200(2.2/3.2) = 137.5 N. Work = 137.5 × 8 = 1100 J.

Question 16

Correct Answer: (D) 24 J

Explanation: v = dS/dt = t². At t=0, u=0; at t=2, v=4 m/s. By Work-Energy Theorem: W = ½m(v² - u²) = ½(3)(16 - 0) = 24 J.

Question 17

Correct Answer: (C) 27 : 256

Explanation: Mass ratio m■/m■ = (r■/r■)³ × (ρ■/ρ■) = (1/8)(3/2) = 3/16. KE■/KE■ = (m■/m■)(v■/v■)² = (3/16)(9/16) = 27/256.

Question 18

Correct Answer: (D) 3 U

Explanation: U■ = ½k(3)² = 9(½k) = U. For total compression 6 cm: U■ = ½k(6)² = 36(½k) = 4U. Increase ΔU = U■ - U■ = 4U - U = 3U.

Question 19

Correct Answer: (A) 50%

Explanation: KE■ = 2.25 KE■. Since momentum p ∝ √KE, p■/p■ = √2.25 = 1.5. Percentage increase = (1.5 - 1) × 100% = 50%.

Question 20

Correct Answer: (B) 3 K

Explanation: Remaining mass m' = (3/4)m, new velocity v' = 2v. New KE K' = ½((3/4)m)(2v)² = (3/4)(4)[½mv²] = 3K.